



**MASTERMIND
INDIA**
Spirit of India Special

Gandhi GOPALAKRISHNAN, 47

INNOVATION

**Eco-friendly bricks
using rice husk ash**

COST

Rs 3.50 a brick

WHAT IT DOES

**Converts ash into bricks,
preventing it from
ruining the environment**

ASHES TO BRICKS

An advocate of natural farming methods for more than a decade, Kanchipuram-based Gandhi Gopalakrishnan was shocked to see lush mango farmland turning infertile by the dumping of rice husk ash (RHA). "The farmers are deceived by claims that RHA helps the fertility of the soil, when the contrary is true," he says. The high pH content (8 per cent) of rice husk ash spells doom for the soil (ideally soil pH should remain between 6.5-7.5 per cent). "In a paddy-growing and processing district like Kanchipuram, I thought of reducing the ecological damage caused by carbon-rich RHA," says Gopalakrishnan, president of the Natural Educational Environment Agricultural Development Society (NEEADS) that he set up in 2004.

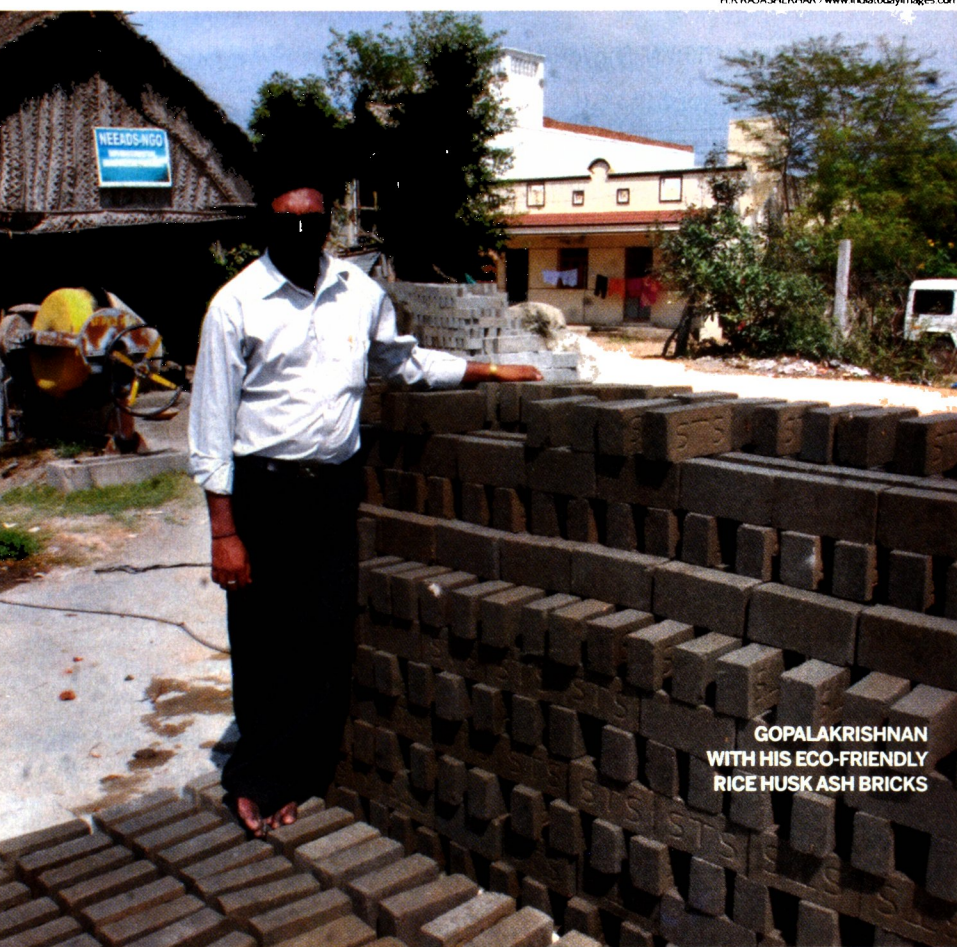
The solution he offered has multiple benefits: convert the hazardous RHA into bricks, preventing it from spoiling the environment. It reduces

the usage of bricks made from top soil, and saves trees too, since mud bricks are solidified by burning with wood. Gopalakrishnan leased out a brick factory, bought a worn-out mosaic machine and started manufacturing RHA bricks, all at a cost of Rs 1 lakh.

The formula is simple: mix RHA with sand, quarry dust and a bit of cement, and an eco-friendly hollow brick is born. In the last two years, NEEADS has manufactured more than 30,000 such bricks. Now they are sold at Rs 3.50 per brick, the same price as a mud brick. "The manufacturing is on a small scale. If we expand the facilities and produce on a mass scale, the price will come down," says Gopalakrishnan. In December 2009, NEEADS received the rural innovation award and a cash price of Rs 2 lakh from Villgro, an organisation that promotes innovations in rural areas.

Having started life as a vendor of vegetable seeds, Gopalakrishnan

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**GOPALAKRISHNAN
WITH HIS ECO-FRIENDLY
RICE HUSK ASH BRICKS**

**THE RHA BRICKS
ALSO REDUCE THE
EROSION OF TOP SOIL
WHICH IS USED IN
MAKING MUD BRICKS.**

today runs an NGO to promote natural farming, which helps around 200 farmers. But his pet project is the eco-friendly bricks. The smoke, which emanates when rice husk is burnt as a means for disposing it, chokes communities and the dumped RHA blocks waterways. "We shouldn't leave a black legacy for future generations. Our children should inherit a cleaner, safer earth. That's at the core of my endeavour," says Gopalakrishnan. Brick by brick, he is laying the foundation of a green future.

by S. Senthil Kumar